



Micro:bit Blocks



How do we give a tiny computer instructions?

A micro:bit is a tiny computer. It does exactly what your code tells it. Let's give it instructions with blocks!

TIME

60 minutes

GROUPS

Pairs (driver and navigator, swap halfway)

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can put blocks in order to make a program.
- ✓ I can find and fix a bug.
- ✓ I can use a sensor to measure something.

STEAM in this activity

S

SCIENCE

A temperature sensor notices heat.

T

TECHNOLOGY

micro:bit and MakeCode block coding.

E

ENGINEERING

Debugging: test, find the mistake, fix it.

A

ART

Design your own 5×5 light picture.

M

MATHS

Counting, variables and a 5×5 grid.



Before the lesson

- ☐ Open MakeCode on each tablet or laptop and try one download before class.
- ☐ Charge or check batteries. One micro:bit, cable and battery pack per pair.
- ☐ Print the pixel grids on this worksheet.
- ☐ Warm up "unplugged": a child is the robot and follows spoken instructions exactly.

Safety

- Hold the micro:bit by its edges.
- Keep the micro:bit and battery pack dry.
- An adult plugs in the cable.



Kit per group

- ☐ A micro:bit and a USB cable
- ☐ MakeCode open on a tablet or laptop
- ☐ MakeCode

Running order (60 minutes)

0–10 min	Unplugged warm-up: program a classmate robot.
10–25 min	Build: smiley face program.
25–40 min	Test 1: button counter. Guess, press 10 times, record.
40–55 min	Test 2: thermometer on the desk and in hands.
55–60 min	Share: who found a bug? How did you fix it?

The investigations

Start here: build: Make a smiley face

Children record: What does the micro:bit show?; Draw what the lights show

What scientists often see: A program is a list of instructions. The computer follows them in order, exactly as written.

Test 1: Button counter: If you press A 10 times, what number will it show?

Children record: What number does it show? *Explorers also:* Draw your code blocks

What scientists often see: A variable is a box that remembers a number. Each press adds 1 to the box. If the number is wrong, programmers look for the bug and fix it.

Test 2: Make a thermometer: Will the micro:bit be warmer on your desk or in your hands?

Children record: What does it show?

What scientists often see: The micro:bit has a sensor that feels how warm it is. Your body is warm, so holding it warms it up. You will use this thermometer in Hot Spots.

Talk about it

- What is a bug? How did you find yours?
- Why does the order of the blocks matter?
- What else could a micro:bit sense?

What to look for

Starter: Drags blocks into the right place and downloads a working program.

Explorer: Predicts what a program will do and explains how they fixed a bug.

Art corner

Pixel picture. Colour squares in the 5×5 grid to design your own picture. Then build it with the "show leds" block.

Maths corner

Starter: Press A 3 times. What number should it show? Draw it.

Explorer: Change the block to "change count by 2". Guess the number after 10 presses: ____.

Real-world link

Programmers test and fix bugs every day. Phones and cars are full of sensors like this one.

Going further

Make the micro:bit show a sad face when the temperature is above 30 °C.

